

Assignment 4: K-Means

In this assignment, you will use a dataset from Kaggle to perform K-means clustering, commonly used for grouping similar data points together. This task will involve several clear steps:

Step 1: Prepare Your Dataset

- Choose a dataset from Kaggle that's good for grouping, like the Mall Customers Dataset ([link](https://www.kaggle.com/vjchoudhary7/customer-segmentation-tutorial-in-python)) (<https://www.kaggle.com/vjchoudhary7/customer-segmentation-tutorial-in-python>). Clean up the dataset by removing any missing or unnecessary data.
- **Attach your data cleaning code** with your report.

Step 2: Visualize Data Before Clustering

- Create a graph to show how your data looks before you apply clustering. This helps you see what the data looks like normally.
- **Attach your plotting code** with your report.

Step 3: Implement K-means Clustering and Determine Optimal Clusters

- Use the K-means algorithm to find groups in your data. To figure out the best number of groups, use the elbow method: plot the within-cluster sum of squares (WCSS) against different numbers of clusters and look for the point where the line starts to flatten out.
- **Attach your K-means and elbow method code** with your report.

Elbow Method Example Code:

```
=====

import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.datasets import make_blobs

# Generate synthetic data
X, _ = make_blobs(n_samples=300, centers=4, cluster_std=0.60, random_state=0)

# Calculate WCSS for different numbers of clusters
```

```
wcss = []
for i in range(1, 11): # Testing 1 to 10 clusters
    kmeans = KMeans(n_clusters=i, init='k-means++', max_iter=300, n_init=10,
random_state=0)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_) # inertia_ is the WCSS for the model

# Plotting the WCSS to observe the 'Elbow'
plt.figure(figsize=(10, 8))
plt.plot(range(1, 11), wcss, marker='o')
plt.title('Elbow Method For Optimal k')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.grid(True)
plt.show()
```

=====

Step 4: Visualize Data After Clustering

- Make another graph to show your data with the groups found by K-means. This shows how the algorithm has organized the data.
- **Attach your post-clustering plotting code** with your report.

Step 5: Interpret Your Results

- Explain what the results from the K-means clustering show about your data. Discuss how this information can help identify specific groups within the dataset.
- **Attach your interpretation and any additional code** you used to analyze the results.

Submission Requirements:

- Submit your completed assignment as a PDF containing all necessary code snippets and graphs.
- Attach corresponding .py and .ipynb files containing executable code.
- Ensure all files are clearly labeled and organized.
- Late submissions will be subject to deductions according to the policy.

Evaluation Criteria: Your assignment will be graded based on the accuracy of your implementation, the clarity of your presentation, and the completeness of your submission. Specific attention will be given to how well you follow these guidelines:

- **Step-by-Step Explanation:** This assignment requires clear, step-by-step explanations. You must explicitly label and explain each step (e.g., Step 1, Step 2, Step 3, Step 4). Failure to provide detailed explanations for each required step will result in a proportional deduction in your score. For example, if the assignment requires four steps and you complete only three, this will result in a score of 75 out of 100.